

SCI-NOI v0.1

SCI-NOI v0.1 Stage B Engineering Conformance Specification

Implementation-blind Stage B draft

SCI-NOI_ENGINEERING_CONFORMANCE v0.1/draft-r0.1

Not owner-accepted or frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Status: implementation-blind draft specification. This document defines what an implementation-facing conformance review would have to demonstrate; it does not report that any implementation conforms.

1. Conformance object

A conformance candidate is one exact tuple:

(implementation identity,
SCI-NOI contract version,
route method and generation,
immutable parent identities,
requested/effective/resolved/applied/realized plans,
product inventory,
evidence generation).

Changing any tuple member creates another candidate. A finite numerical payload does not establish conformance when identity, lifecycle, or required states are missing. Review of one method, product, persistence mode, observation, array, or execution lane does not imply another.

2. Required plan record

The GEN plan record shall expose, without relying on a hidden default:

- route method/version and fixed versus relearned class;
- exact earliest parent and PTC-to-MAP insertion boundary;
- exact parent coefficient, projection, support, response, WCS, validity, and lifecycle identities;
- canonical observation, detector, array, group, and readout-network IDs;
- exact rational tau_h for every observation/network stratum;
- exact random-bit generator content identity and algorithm version;
- opaque seed/key bytes, key namespace, canonical serialization version, and counter-domain rules;
- B_requested and positive retry cap;
- persistence mode plus requested/effective/applied/realized states;
- terminal failure policy; and
- exact profile/source versions for every named admission decision.

Missing or conflicting fields make the candidate unavailable before numerical execution. This is the engineering projection of NOI-REQ-002 through NOI-REQ-011, NOI-REQ-017, and NOI-REQ-029 through NOI-REQ-034.

3. Deterministic design construction

For each canonical observation/network stratum, a conforming design builder shall:

1. construct the stable detector domain from scientific identities, never encounter order;
2. derive every positive B_d from exactly the frozen admitted parent

contributions;

3. represent each persisted contribution and τ_h as an exact rational and evaluate the integer/rational cross-multiplied predicate without an order-dependent reduction;

4. obtain independent uniform candidate bits from disjoint canonical member/stratum/candidate-counter key domains;

5. map bit 0 to -1 and bit 1 to +1;

6. accept the first candidate satisfying the stratum predicate;

7. fail the complete design if a required stratum exhausts its positive retry cap; and

8. compose accepted stratum vectors in canonical observation/stratum/unit order.

Candidate rejection shall not create a member identity. No builder may relax a tolerance, balance across strata, alter a parent population, force a complement, drop a duplicate, or silently change RNG, cap, count, or persistence mode.

The target-law statement is scientific, while any empirical demonstration of generator behavior belongs to a later evidence plan. This draft makes no such demonstration.

4. Design publication and exact reconstruction

A successful design publication shall include:

- canonical ordered coherence domain and digest;
- exact mass inputs or their content-bound reconstructing parent identity;
- every tolerance and exact comparison representation;
- generator/version, seed/key namespace, and key/counter derivation;
- assignment-design generation and each member identity;
- exact assignment bytes or the complete deterministic reconstruction record;
- assignment digest followed by byte-equality duplicate confirmation;
- complement-orbit relation;
- all distinct member counts;
- the uncentered member-by-coherence sign matrix rank and null-space record;
- equal weight $1/B_{\text{resolved}}$ for every successfully resolved member; and
- requested/effective/resolved/applied/realized lifecycle states.

Reconstruction shall be invariant to worker count, process scheduling, traversal order, container layout, and writer arrival. A digest verifies identity but does not reconstruct omitted values. Compact regeneration is conforming only when the complete record reconstructs the declared product under its stated byte-identical or numerical reproducibility class.

5. Member production and atomic failure

For every admitted member, the producer shall bind the exact assignment to the exact admitted PTC occurrences and apply it before identical frozen MAP arithmetic. Inline and materialized representations are equivalent only under that equality. The producer shall publish exact parent/operator, unit/beam, WCS, support, response, source-imprint, QC, persistence, lifecycle, terminal state, cause, and provenance facts.

An admitted-member failure fails the whole ensemble for UNC. A conforming implementation shall not publish completed survivors or a partial streaming accumulator as an UNC-authorized ensemble. Diagnostic retention shall remain bound to the failed ensemble. Retry or replacement requires another exact generation.

6. UNC construction

Before calculation, the exact member policy shall evaluate every member and the exact ensemble policy shall evaluate the complete design. Only the named positive conjunction authorizes the named action; producer completion facts remain producer-owned.

For the initial estimator, a conforming calculation shall:

- construct D_{common} by exact identity over every admitted member;
- reject survivor, pairwise, zero-fill, interpolation, and domain-extension

substitutes;

- accumulate $\text{sum}_b (1/B_{\text{resolved}}) M_b(p)^2$ without empirical recentering or

a B-1 correction;

- preserve squared signal units and the exact source-bearing conditional

randomization label;

- publish dependence, counts, rank, effective-information, and estimator-

uncertainty state; and

- publish an atomic product or a typed unavailable/failed state.

For the reciprocal, the implementation shall construct only the finite strictly positive subdomain and preserve unavailability elsewhere. No zero, floor, cap, clipping, epsilon, shrinkage, inverse-variance, precision, or MAP coefficient interpretation is conforming under this method identity.

7. STD construction

The standardization admission decision shall bind the immutable normalized real-observation MAP numerator to the exact compatible $V_{\text{hat_cond}}$ generation from the same frozen operator state. A conforming calculation shall use the canonical square root, exact compatible finite-positive intersection, unit 1, and the exact claim in NOI-REQ-028. It shall publish unavailable rather than zero or infinity on invalid divisions. JINC, an algebraic inverse-scale route, interpolation, substituted response, or another generation requires a separate method.

8. External transformations, Wiener, and FRUIT

Conformance for a transformed route requires exact owner authority and parity for purpose, algorithm, state, parameters, operation order, domain, support, edge and missing-data behavior, normalization, unit, response, validity, lifecycle, and failure semantics. Applying a transform with a similar name or numerical output is insufficient.

If the transform is learned or updated from a prior NOI product, the candidate shall publish distinct immutable prior-input, owner-learning, transformation, science-product, GEN, and successor-UNC generations plus their dependence. A prior NOI product shall not be presented as independent validation. Per-member learning or FRUIT replay is a different relearned method and cannot share a fixed ensemble.

9. Static conformance checks required before numerical evidence

A later conformance review shall first establish all of the following without executing a scientific reduction:

- every identifier and method version is exact and collision-free;
- the ordinary route is the selected PTC-to-frozen-MAP method;
- all unavailable parent gates remain explicit;
- no default is supplied for tolerance, RNG, seed/key, retry cap, count, or

persistence mode;

- canonical keys contain every required field in the required order;
- all counts, duplicate/complement states, rank, and weights are separately

represented;

- no partial-success state can project to an UNC action;
- UNC and STD formulae, domains, units, and unavailable behavior are exact;
- every profile projects only its named action;
- parents and prior generations are immutable; and
- forbidden claim labels are absent from realized-success semantics.

Passing static checks would establish only algebraic/structural consistency of the candidate representation. It would not establish implementation conformity, numerical fidelity, calibration, physical-noise meaning, covariance completeness, significance, performance, readiness, or production authority.

10. Requirement and prediction traceability

The normative source of requirements is NORMATIVE_CORE.md. The analytic prediction source is SCIENTIFIC_RATIONALE.md. Exact row-level mapping is in REQUIREMENT_PREDICTION_TRACEABILITY.csv. The package-local verifier requires every NOI-REQ-001 through NOI-REQ-037 and every NOI-PRED-001 through NOI-PRED-015 to appear in that crosswalk, rejects duplicate IDs, verifies that every cited authority code is one of the 17 manifest-admitted objects, and verifies the deterministic source/PDF build manifest.

The crosswalk is a documentation trace, not evidence that a prediction has been tested or a requirement implemented.

11. Claim ceiling

This specification defines a future review target only. It makes no implementation-conformity, validation, calibration, physical-noise, covariance-completeness, Gaussian-significance, achieved-performance, readiness, freeze, production-suitability, or production-authorization claim.

Appendix A. Exact Traceability Rows

NOI-REQ-001

Predictions: NOI-PRED-015 Admitted authorities: SCOPE|COVER|TAX|LIFE Engineering sections: 1|10
Review kind: identifier_and_role_review

NOI-REQ-002

Predictions: NOI-PRED-007 Admitted authorities: OWNER|PTCB|GRAPH|MAPB Engineering sections: 1|2|5
Review kind: exact_parent_graph_review

NOI-REQ-003

Predictions: NOI-PRED-014 Admitted authorities: OWNER|CONV|GRAPH|FILTER Engineering sections: 1|8
Review kind: method_graph_review

NOI-REQ-004

Predictions: NOI-PRED-002 Admitted authorities: OWNER|PTCB|DESIGN Engineering sections: 2|3 Review
kind: canonical_domain_review

NOI-REQ-005

Predictions: NOI-PRED-003|NOI-PRED-004 Admitted authorities: OWNER|PTCB|DESIGN Engineering sections:
2|3 Review kind: exact_mass_and_stratum_review

NOI-REQ-006

Predictions: NOI-PRED-001|NOI-PRED-003|NOI-PRED-004 Admitted authorities: OWNER|DESIGN Engineering
sections: 2|3 Review kind: exact_predicate_review

NOI-REQ-007

Predictions: NOI-PRED-001|NOI-PRED-002 Admitted authorities: OWNER|DESIGN|CORE1 Engineering
sections: 3 Review kind: target_law_review

NOI-REQ-008

Predictions: NOI-PRED-006|NOI-PRED-007 Admitted authorities: OWNER|DESIGN|CORE1|CORE2 Engineering
sections: 3|4 Review kind: joint_design_review

NOI-REQ-009

Predictions: NOI-PRED-005 Admitted authorities: OWNER|DESIGN|CORE1 Engineering sections: 2|3 Review
kind: plan_completeness_review

NOI-REQ-010

Predictions: NOI-PRED-005 Admitted authorities: DESIGN|CORE1 Engineering sections: 3|4 Review kind:
canonical_serialization_review

NOI-REQ-011

Predictions: NOI-PRED-003|NOI-PRED-005 Admitted authorities: OWNER|DESIGN Engineering sections: 3
Review kind: fail_closed_design_review

NOI-REQ-012

Predictions: NOI-PRED-006|NOI-PRED-007 Admitted authorities: DESIGN|CORE1|CORE2 Engineering sections: 4 Review kind: count_and_equivalence_review

NOI-REQ-013

Predictions: NOI-PRED-006|NOI-PRED-013 Admitted authorities: DESIGN|UNCT|CORE2 Engineering sections: 4|6 Review kind: rank_and_information_review

NOI-REQ-014

Predictions: NOI-PRED-006|NOI-PRED-009 Admitted authorities: OWNER|DESIGN|UNCT Engineering sections: 4|6 Review kind: weight_identity_review

NOI-REQ-015

Predictions: NOI-PRED-010 Admitted authorities: OWNER|DESIGN|CORE1|COVER Engineering sections: 5 Review kind: source_imprint_review

NOI-REQ-016

Predictions: NOI-PRED-008 Admitted authorities: OWNER|DESIGN|LIFE|VALP Engineering sections: 5|6 Review kind: atomic_completion_review

NOI-REQ-017

Predictions: NOI-PRED-008 Admitted authorities: OWNER|DESIGN|LIFE|VALP Engineering sections: 2|4|5 Review kind: persistence_mode_review

NOI-REQ-018

Predictions: NOI-PRED-014 Admitted authorities: OWNER|CONV|GRAPH|FILTER Engineering sections: 8 Review kind: transformation_parity_review

NOI-REQ-019

Predictions: NOI-PRED-014 Admitted authorities: OWNER|CONV|GRAPH|FILTER|LIFE Engineering sections: 8 Review kind: successor_generation_review

NOI-REQ-020

Predictions: NOI-PRED-009|NOI-PRED-010|NOI-PRED-013 Admitted authorities: UNCT|CORE1|CORE2 Engineering sections: 6 Review kind: unc_method_identity_review

NOI-REQ-021

Predictions: NOI-PRED-008 Admitted authorities: OWNER|UNCT|VALP|LIFE Engineering sections: 6 Review kind: unc_admission_review

NOI-REQ-022

Predictions: NOI-PRED-009|NOI-PRED-010 Admitted authorities: OWNER|UNCT|CORE2 Engineering sections: 6 Review kind: estimator_formula_review

NOI-REQ-023

Predictions: NOI-PRED-009 Admitted authorities: OWNER|UNCT|VALP Engineering sections: 6 Review kind: common_domain_review

NOI-REQ-024

Predictions: NOI-PRED-010|NOI-PRED-013 Admitted authorities: OWNER|UNCT|CORE1|CORE2 Engineering sections: 6 Review kind: claim_and_metadata_review

NOI-REQ-025

Predictions: NOI-PRED-014 Admitted authorities: OWNER|UNCT|CORE1|CORE2 Engineering sections: 6|8 Review kind: covariance_representation_review

NOI-REQ-026

Predictions: NOI-PRED-011 Admitted authorities: OWNER|UNCT|LIFE Engineering sections: 6 Review kind: inverse_domain_review

NOI-REQ-027

Predictions: NOI-PRED-012 Admitted authorities: OWNER|STDT|MAPB|VALP Engineering sections: 7 Review kind: std_formula_and_join_review

NOI-REQ-028

Predictions: NOI-PRED-012 Admitted authorities: OWNER|STDT|COVER|LIFE Engineering sections: 7 Review kind: std_unit_and_claim_review

NOI-REQ-029

Predictions: NOI-PRED-015 Admitted authorities: OWNER|VALP|CONV Engineering sections: 2|6|7|10 Review kind: profile_state_review

NOI-REQ-030

Predictions: NOI-PRED-015 Admitted authorities: OWNER|VALP|PTCB Engineering sections: 2|5 Review kind: generation_action_review

NOI-REQ-031

Predictions: NOI-PRED-008|NOI-PRED-015 Admitted authorities: OWNER|VALP|LIFE Engineering sections: 5|6 Review kind: member_action_review

NOI-REQ-032

Predictions: NOI-PRED-008|NOI-PRED-009|NOI-PRED-011|NOI-PRED-015 Admitted authorities: OWNER|VALP|UNCT Engineering sections: 6 Review kind: ensemble_action_review

NOI-REQ-033

Predictions: NOI-PRED-012|NOI-PRED-015 Admitted authorities: OWNER|VALP|STDT Engineering sections: 7 Review kind: standardization_action_review

NOI-REQ-034

Predictions: NOI-PRED-003|NOI-PRED-008|NOI-PRED-009|NOI-PRED-011|NOI-PRED-012 Admitted authorities: CONV|MAPB|JINCB|PTCB|LIFE Engineering sections: 1|2|3|5|6|7 Review kind: typed_unavailability_review

NOI-REQ-035

Predictions: NOI-PRED-014|NOI-PRED-015 Admitted authorities: CONV|MAPB|JINCB|PTCB|FILTER|LIFE Engineering sections: 1|8 Review kind: immutability_review

NOI-REQ-036

Predictions: NOI-PRED-013 Admitted authorities: OWNER|CONV|UNCT|LIFE Engineering sections: 6|8
Review kind: role_nonpromotion_review

NOI-REQ-037

Predictions: NOI-PRED-001|NOI-PRED-002|NOI-PRED-003|NOI-PRED-004|NOI-PRED-005|NOI-PRED-006|NOI-PRED-007|NOI-PRED-008|NOI-PRED-009|NOI-PRED-010|NOI-PRED-011|NOI-PRED-012|NOI-PRED-013|NOI-PRED-014|NOI-PRED-015 Admitted authorities: SCOPE|COVER|OWNER|CONV|MAPB|JINCB|PTCB|GRAPH|DESIGN|UNCT|STDT|VALP|FILTER|LIFE|CORE1|CORE2
Engineering sections: 9|10|11 Review kind: claim_ceiling_review